

Sarah Binta Alam Shoilee

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🔄 Shoilee

in sarahshoilee



Education

- 2021 – ongoing 📖 **Ph.D. in Computer Science, Vrije Universiteit Amsterdam**, the Netherlands.
Thesis Title: *Knowledge Discovery for Provenance Research on Colonial Heritage Objects*.
– In my PhD, I explored various stages of the knowledge discovery process through a series of studies addressing different challenges—from modeling polyvocal narratives and aligning long-tail entities: sparse, ambiguous data to discovering knowledge in skewed datasets using semantic relations and communicating through visualization. Currently, I am working on extracting military career events from historical tabular image records to build enriched knowledge graphs.
– *Expected to finish by the end of November, 2025*
Promoters: Prof. Dr. Jacco van Ossenbruggen & Prof. Dr. Susan Legêne
Co-promoter: Dr. Victor de Boer
- 2018 – 2020 📖 **M.Sc. in Artificial Intelligence, KU Leuven**, Belgium.
Specialization: Big Data Analytics
Thesis Title: *Knowledge Discovery on Spreadsheet using Semantic Information*.
– Developed Semantic-TaCLE, a heuristic-driven system for spreadsheet formula reconstruction using semantic cues from tabular headers.
Thesis Supervisor: Prof. Dr. Luc De Raedt
- 2016 – 2017 📖 **M.Sc. in Computer Science & Engineering, North South University**, Bangladesh.
🏆 *Summa Cum Laude*
- 2012 – 2016 📖 **B.Sc. in Computer Science & Engineering, North South University**, Bangladesh.

Teaching and Supervision Experience

- 2022 – 2024 📖 **Thesis Supervisor, Vrije Universiteit Amsterdam**, the Netherlands.
– Independently supervised one Bachelor thesis in Computer Science.
– Co-supervised two Master's theses in Information Studies, one in Artificial Intelligence and one Bachelor project in Artificial Intelligence.
- 2021 – 2023 📖 **Teaching Assistant, Vrije Universiteit Amsterdam**, Bangladesh.
Grading assistance for the course: Knowledge Organization
- 2017 – 2018 📖 **Teaching Assistant, North South University**, Bangladesh.
Instructed and conducted practical sessions for undergraduate students in courses: Database Management, Digital Logic Design and Web Development

Pre-PhD Research Experience (Interdisciplinary)

- 2020 📖 **UX Research Trainee, Hogeschool van Amsterdam**, the Netherlands.
Designed and evaluated user-interface for an explainable AI-driven web platform for promoting healthy lifestyles, through user-centered studies in collaboration with TNO.
- 2017 – 2018 📖 **Research Assistant, North South University**, Bangladesh.
– Conducted focus groups and qualitative analysis for *Google's Next Billion Users Initiative*, proposing solutions to enhance product adoption among tech users in Bangladesh.
– Designed and evaluated a mobile application interface for breast cancer awareness, focusing on user satisfaction and sustainable adoption.

Skills and Competencies

Programming Languages	📖 Python , C/C++, Java, Node.js Proficient in scripting, data analysis, and backend development.
Web & Data Technologies	📖 HTML, CSS, JavaScript, SQL Experience with front-end prototyping
Semantic Web Technologies	📖 SPARQL , SHACL, RDF, OWL, Data Modeling, KG Constructions Skilled in data retrieval, management, and integration for structured and unstructured data.
Machine Learning Techniques	📖 Computer Vision, Natural Language Processing, Data Mining Theoretical and practical understanding of different approaches.
Tools & Frameworks	📖 Git , Docker, $\LaTeX$ Familiar with collaborative tools and semantic annotation environments.
Languages	📖 English (fluent), Bangla (native), Dutch (basic)
Current Research Focus	📖 Linked Open Data, Interdisciplinary Research, Knowledge Discovery, Digital Humanities

Research Publications

Conference Proceedings

- 1 S. B. A. Shoilee, A. Ahola, H. Rantala, *et al.*, “Enhancing provenance research with linked data: A visual approach to knowledge discovery,” in *Proceedings: SemDH 2025 Second International Workshop of Semantic Digital Humanities, co-located with ESWC 2025, Portoroz, Slovenia*, 🏆 Best Paper, CEUR Workshop Proceedings, 2025.
- 2 S. B. A. Shoilee, A. Ahola, H. Rantala, *et al.*, “Pm-sampo: Semantic portal for heritage object provenance research,” in *The Semantic Web: ESWC 2025 Satellite Events, Portoroz, Slovenia, June 1 - 5, 2025, Proceedings*, ★ Nominated for Best Demo, Springer-Verlag, 2025.
- 3 S. B. A. Shoilee, V. de Boer, and J. van Ossenbruggen, “A framework for evaluating entity alignment impact on downstream knowledge discovery,” in *International Conference on Knowledge Engineering and Knowledge Management*, Springer, 2024, pp. 226–242.
- 4 S. B. A. Shoilee, V. de Boer, and J. van Ossenbruggen, “Polyvocal knowledge modelling for ethnographic heritage object provenance,” in *Knowledge graphs: Semantics, machine learning, and languages*, IOS Press, 2023, pp. 127–143.
- 5 S. B. A. Shoilee, “Knowledge discovery for provenance research on colonial heritage objects,” in *Proceedings: Doctoral Consortium at ISWC 2022 co-located with 21st International Semantic Web Conference (ISWC 2022)*, CEUR Workshop Proceedings, 2022.

Journal Articles

- 1 B. Sartini, S. B. Alam Shoilee, C. A. Libbi, and V. De Boer, “Multivocal exhibition: Exploring cultural perspectives through user-curated art exhibitions,” *ACM Journal on Computing and Cultural Heritage*, vol. 17, no. 4, pp. 1–21, 2025.

Books and Chapters

- 1 V. de Boer and S. B. A. Shoilee, “Linked open data for cultural heritage,” in *he Palgrave Encyclopedia of Cultural Heritage and Conflict*, I. Saloul and B. Baillie, Eds., Cham: Publisher Name, 2025, pp. 1–7.

Miscellaneous Experience

Awards and Achievements

- 2025  **Best Paper Award**, Second International Workshop of Semantic Digital Humanities (SemDH 2025), co-located with ESWC-2025.
-  **Nomination for Best Demo**, in Extended Semantic Web Conference (ESWC-2025).
- 2022  **Polyvocality Award**, HackaLOD 2022, Utrecht, the Netherlands.
- 2018  **SIGCHI Travel Grant**, ACM SIGCHI and HCI Professionals Association of India.
- 2017  **Merit Excellence Award**, North South University, Dhaka, Bangladesh.

Research Visits

- Jan-Mar 2025  **Research Visit, Aalto University**, Finland.
Collaborated with SeCo Group, lead by *Prof. Eero Hyvönen*, to develop a prototype of the PM-Sampo, semantic web application for cultural heritage knowledge discovery.

Hackathons

- November 2022  **Participant, HackaLOD**, Utrecht, the Netherlands.
Participated with a team of PhD researchers from the Cultural AI Lab in a national Linked Data hackathon on cultural heritage and developed a multi-vocal exhibition for museum using Linked Data storytelling.

Summer Schools

- July 2022  **International Semantic Web Research Summer School (ISWS)**, Bertinoro, Italy.
- May 2018  **SIGCHI Summer School: Research Methods and Approaches to Text Entry and Other Interaction Techniques**, IIT Bombay, Mumbai, India.

Invited Talks

- February 2024  **Polyvocal Knowledge Modelling for Ethnographic Heritage Object Provenance.**
Cultural AI Event — Open Up Your Collections!
- January 2024  **Guest Lecture, Vrije University Amsterdam**
Course: Digital Humanities and Social Analytics in Practice.
- July 2023  **Talk: Tracing the Unknown: Learning from Provenance Data.**
Conference: Museums, Collections and Society.
- April 2022  **Graph Mining on Historical Past to Digitally Trace Colonial Objects.**
Create Salon at UvA — Polyvocal Perspectives in Archives.

Peer Review Activities

- 2022 – 2025  **Reviewer.**
Conferences: ISWC (2022, 2023, 2024), EKAU (2024), SemDH Workshop (2024, 2025).
Journal: Journal on Computing and Cultural Heritage (JOCCH).

Organizational Activities

- 2023 – 2025  **Coordinator and Host, Cultural AI Reading Club**, the Netherlands.
Organized and moderated monthly interdisciplinary reading sessions, facilitating cross-domain scholarly engagement and discussions.
- 2024  **Volunteered for SEMANTICS-2024 Conference Organization**, the Netherlands.
- 2021  **Volunteered for SEMANTICS-2021 Conference Organization**, the Netherlands.

References

Available on Request